ML

Gaussian Assumption $y(t) \sim N(0, R)$

likelihood function $L(R) = \prod_{l=1}^{L} p(y_l | R) = \left(\frac{1}{\pi^M \det(R)} \right)^L e^{-\sum_{l=1}^{L} y_l^H R^{-1} y_l}$

$\log L(R) = -\mathrm{tr}(Y^H R^{-1} Y) - L \log\det(R) - ML \log \pi$

$R_{ML} = \arg\max \log L = \arg\min \ \mathrm{tr}(\hat{R} R^{-1}) + \log\det(R)$

No need P diagonal

	P diagonal	A ULA	V uniform	
COMET	×	×	×	only criteria
SPICE	✓	×	×	SDP, iterative, SOPP
SPICE-ML	✓	✓	×	iterative closed-form
Toeplitz AML	✓	✓	✓	directly closed-form

$\hat{r} = \mathrm{vec}(\hat{R})$, $r = \mathrm{vec}(R)$

$M^2 \times 1$

$\hat{\gamma} = J\,\mathrm{vec}(\hat{R})$ real

$M^2 \times 1$

$\hat{R} = \begin{bmatrix} z_1 & \bar{z}_2 \\ z_2 & z_3 \end{bmatrix}$, $\hat{r} = \begin{bmatrix} z_1 \\ z_2 \\ \bar{z}_2 \\ z_3 \end{bmatrix}$. $J = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 1/2j & 1/2j & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. $\hat{\gamma} = \begin{bmatrix} z_1 \\ \mathrm{Re}(z_2) \\ \mathrm{Im}(z_2) \\ z_3 \end{bmatrix}$

EXIP: assume $\hat{\alpha} = f(\hat{\theta})$ and $\tilde{\alpha}$ have same limitation

$D_\alpha \subset \tilde{D}_\alpha$, then

$\tilde{\theta} := \arg\min \ (\tilde{\alpha} - f(\theta))^T L''(\tilde{\alpha}) (\tilde{\alpha} - f(\theta))$

has $\tilde{\theta} \to \hat{\theta}_{ML}$ asymptotically

$\theta, \alpha : R$ $\quad f$: Identity map

D_α: Toeplitz space $\quad \tilde{D}_\alpha : \mathbb{R}^{M \times M}$

$\hat{\theta}_{ML}, \hat{\alpha}_{ML} : \hat{R}$ by directly solving ML

$\tilde{\alpha} : \hat{R}$ since $\hat{R} \to R$

$\tilde{\theta}$: COMET solution R_{COMET}

Covariance Matching / Weighted LS / Generalized LS

$\hat{\gamma} = f(\theta, P, \sigma^2)$,

unconstraint ML

$\tilde{\gamma} = \hat{r}$ without Toeplitz structure

$\nabla_R \log|R| + tr(\hat{R}R^{-1}) = 0$

since $\hat{r}$ is ML solution, $\hat{r} \to r$

$R^{-T} - R^{-T}\hat{R}R^{-T} = 0$

$R_{ML} = \hat{R}$

$\nabla \log\det(X) = X^{-T}$

$\tilde{\phi} = (\tilde{\theta}, \tilde{P}, \tilde{\sigma}^2) = \arg\min\ (\hat{\gamma} - f(\phi))^T \Gamma^{-1} (\hat{\gamma} - f(\phi))$

$\nabla tr(AX^{-1}) = -X^{-T}AX^{-T}$

$\min\ (\gamma - \hat{\gamma})^T Cov^{-1}(\gamma)(\gamma - \hat{\gamma})$ Weighted LS

$L''(R)$ 不会算，不知道为什么直接用 $Cov(\gamma)$

Assumption circularly-symmetric Gaussian : Covariance Hermitian

$Cov(r) = \frac{1}{L}(R^T \otimes R)$ 按元素展开推导

$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{bmatrix}$

ML achieve CRLB asymptotically $Cov(\hat{r}) \to I^{-1}(r)$

Fisher Information Matrix $I(\phi)_{ij} = tr\left(\frac{\partial R}{\partial \phi_i} R^{-1} \frac{\partial R}{\partial \phi_j} R^{-1}\right)$

$= \left[\frac{\partial r}{\partial \phi_i}\right]^H (R^{-T} \otimes R^{-1}) \left[\frac{\partial r}{\partial \phi_j}\right]$?

raplace $Cov(r)$ by $Cov(\hat{r})$ (by EXIP)

$\min\ (r - \hat{r})^H (\hat{R}^{-T} \otimes \hat{R}^{-1})(r - \hat{r})$

$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$

$(A \otimes B)^H = A^H \otimes B^H$

$f = \| \hat{R}^{-\frac{1}{2}} (R - \hat{R})^{-1} \hat{R}^{-\frac{1}{2}} \|_F^2 = tr[(\hat{R} - R)\hat{R}^{-1}(\hat{R} - R)\hat{R}^{-1}]$

$vec(ABC) = (C^T \otimes A)\, vec(B)$

if B is diagonal, $vec(ABC) = (C^T \odot A)\, diag(B)$

Khatri-Rao product $A \odot B = (a_1 \otimes b_1 \ \cdots \ a_N \otimes b_N)$